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REVIEW

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Food allergy in a changing dietary landscape: A focus on the Asia Pacific region

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Abstract

Recent studies have provided compelling evidence to suggest that various environmental factors play a significant role in the development of food allergies. As our society experiences rapid economic growth, increased urbanization, and a shift towards a more Westernized diet, the incidence of food allergies is also on the rise and the pattern is gradually evolving. This review will delve into the changes in the epidemiology of food allergies within the Asia-Pacific region and the various dietary practices and factors that are postulated to play a role in the rise in food allergies over the years. Although there have been important advancements in the field of food allergies, there are still numerous uncertainties regarding the intricate relationship between diet and food allergies. Specifically, the role of epigenetic factors in influencing the susceptibility to food allergies, as evidenced by studies that assessed the impact of migration and rural–urban dynamics, is not fully understood. Addressing this knowledge gap presents an opportunity to develop more effective prevention and treatment strategies that could greatly benefit individuals living with food allergies.

KEYWORDS
diet, epidemiology, food allergy, nutrition transition

Abbreviations: ALADDIN, assessment of lifestyle and allergic disease during infancy; AP, Asia-Pacific; APAPARI, Asia Pacific Academy of Pediatric Allergy, Respirology & Immunology; FPIES, Food Protein-Induced Enterocolitis Syndrome; HK, Hong Kong; JSPACI, Japanese Society of Pediatric Allergy and Clinical Immunology; MHLW, Ministry of Health, Labour and Welfare; PARSIFAL, prevention of allergy risk factors for sensitization in children related to farming and anthroposophic lifestyle; PETIT, prevention of egg allergy with tiny amount of intake; PIFA, prevalence of infant food allergy; SCFAs, short-chain fatty acids; sIgE, specific IgE; SPT, skin prick test; US, United States; UPFs, ultra-processed foods.

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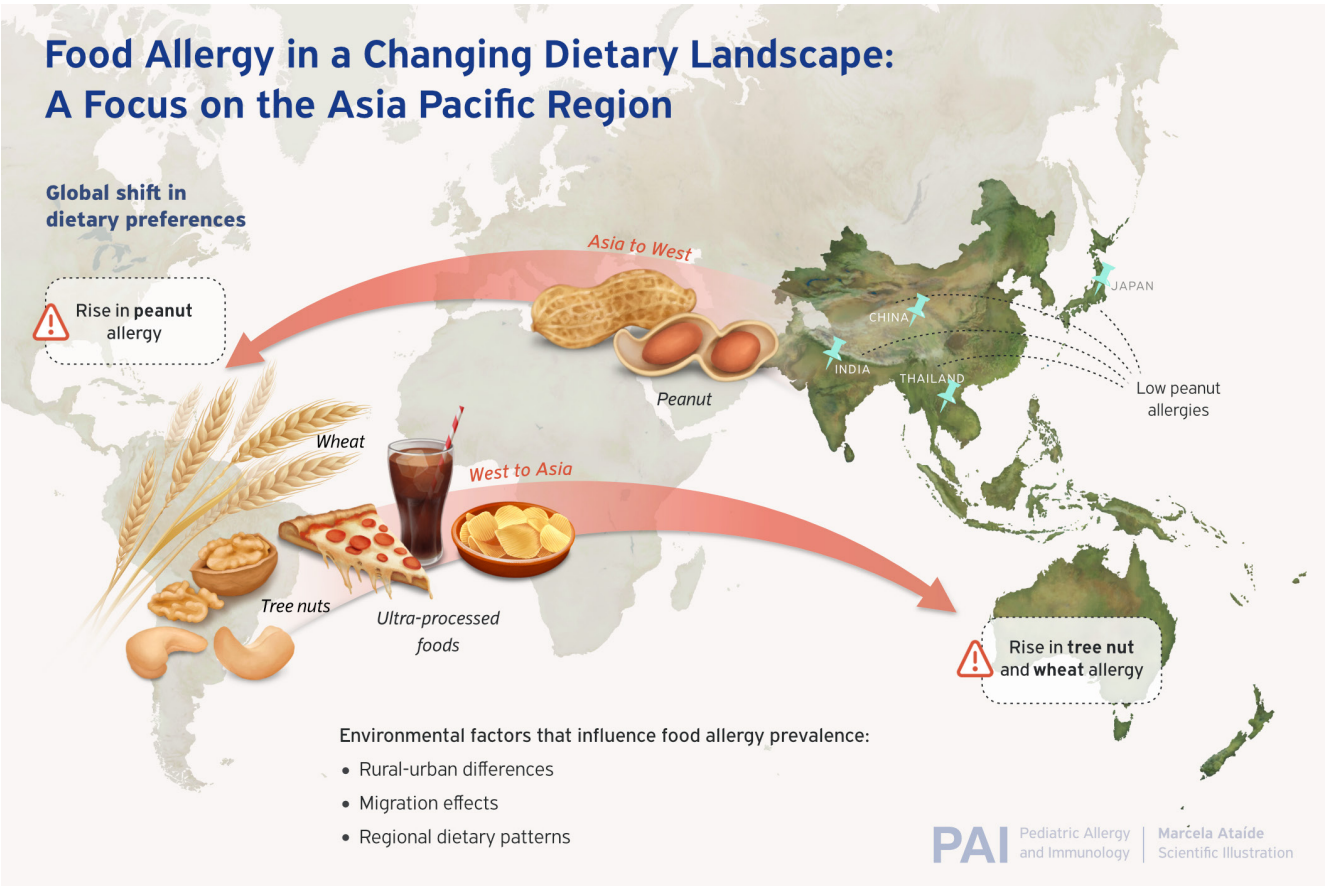

GRAPHICAL ABSTRACT

1 | INTRODUCTION

Food allergy prevalence has been steadily rising in various regions of the world,[1] with the rise most evident in developed countries.[2] However, the pattern of food allergy differs based on geographical region, culture, and diet,[3] which are elements of the "exposome." The exposome is defined as the culmination of an individual's lifetime exposures and how those exposures impact health.[4] The interaction between environmental exposures and individuals' unique features, such as genetics, is recognized as an increasingly important area of research as genetics alone cannot explain the increase in food allergy prevalence in the past few decades.[5,6] The human diet is an essential exposome and the nutritional transition to more Westernized patterns of dietary intake is one of the most remarkable effects of global urbanization.[7]

The Asia Pacific (AP) region is a vast area that is home to a significant portion of the world's population. It encompasses East Asia (including countries such as China, Japan, and Korea), Southeast Asia (Vietnam, Thailand), South Asia (India, Sri Lanka), and Oceania (Australia, New Zealand). The AP region has diverse populations with varied economic development, dietary preferences, and environmental exposures. In contrast, Western countries are often referred to as the economically developed nations of North America and Western Europe with high levels of industrialization. By studying and understanding their differences in prevalence and patterns of food allergies, researchers can gain insights into how these factors influence the development and expression of allergies.

This review aims to examine the pattern of food allergy in the AP Region, compared with the West, and analyze changes in food allergy over time and potential links with changes in the global

Key message

The rise in the prevalence of food allergies can largely be attributed to environmental factors, which are influenced by the adoption of Westernized lifestyles and diets. The East's historic dietary pattern contrasts with the West's, highlighting potential protective dietary features. The shift in the pattern of peanuts, tree nuts, wheat, and egg allergies calls for ongoing research and public health strategies to mitigate their impact. There is a need for a better understanding of how different dietary patterns affect allergic diseases.

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patterns of dietary intake. The goal is to gain insights into the role of diet in the prevalence of food allergy.

2 | EPIDEMIOLOGICAL TRANSITION OF FOOD ALLERGY

2.1 | Time-trend prevalence of food allergy

The prevalence of food allergy is widely believed to have increased in developed countries over the past decades, although relatively few studies have measured changes in food allergy in the same population over time using objective, gold-standard measures of food allergy.[1] Changes in the incidence of food-induced anaphylaxis over time provide a proxy measure of IgE-mediated food allergy prevalence, and a systematic review in 2019 found an increase in total or food-induced anaphylaxis incidence over time reported by 19 studies, with studies in developing countries underrepresented.[8]

More recent studies suggest that the prevalence of IgE-mediated food allergy may be stabilizing, although data remain sparse. In three cross-sectional surveys conducted in China, each including between 300 and 500 children, the prevalence of food allergy increased significantly from 3.5% to 7.7% between 1999 and 2009, with a smaller, non-significant increase between 2009 and 2019.[9] Three cohort studies conducted on the Isle of Wight, UK, showed an increase in peanut allergy between children born in 1989 and those born between 1994 and 1996 from 0.5% to 1.4%, with no further increase in children born between 2001 and 2002.[10] Two cohort studies in Melbourne, Australia also showed that the prevalence of peanut allergy remained stable at around 3% between 2007–2011 and 2018–19 (Table 1).[11] It remains unclear why there has been a stabilization in the prevalence of food allergy over the past decade. Efforts to promote the early introduction of allergenic foods starting around 2016 may have contributed to this stabilization in some countries, although they may not account fully for the above observations. In addition, changes in dietary and environmental factors may have reached a plateau or started to reverse, contributing to stabilization in food allergy prevalence.

There are limited time-trend prevalence data on non-IgE-mediated food allergies. However, Food Protein-Induced Enterocolitis Syndrome (FPIES) is being increasingly reported in the United Kingdom, United States, Spain, Israel, and Japan,[12,13] with an apparent increase in egg and peanut-induced FPIES in Japan and the United States, respectively.[14–16] This will be further discussed in the next section.

2.2 | Rural–urban differences in the prevalence of food allergy

Substantial evidence from recent studies has shown that children who grow up on farms are less likely to develop allergies and asthma than children who live in urban areas.[17] In the context of food allergy, the EuroPrevall INCO group, which consists of centers from China, India, and Russia, undertook a cross-sectional survey on food allergy prevalence as the second phase of the multicenter EuroPrevall survey.[18] The study recruited 35,549 children aged 6 to 11 years from China (Hong Kong [HK], Guangzhou & Shaoguan), Russia (Tomsk), and India (Mysore & Bengaluru). The prevalence of probable food allergy was determined by reported symptoms and positive skin prick test (SPT) or specific IgE (sIgE) (≥0.7 kUA/L) to the specific food. Notably, despite a similar genetic background, the prevalence of probable food allergy was significantly higher in HK (1.5%), a highly Westernized city, compared to the two other Chinese centers, Shaoguan (0.69%) and Guangzhou (0.21%), which still retain some traditional lifestyle elements. Children born in HK were four times more likely to develop food sensitization than those born in mainland China who migrated to HK.[18]

In a cross-sectional study conducted in Vietnam on 4443 children between the ages of 2–6 years in Hue City and 4177 children in Tien Giang Province,[19] children in Hue City had a higher prevalence of doctor-diagnosed food allergies (8.4%) than those in Tien Giang Province (4.96%). Similarly, the South African Food Allergy study showed a higher prevalence of food sensitization and allergy in urban children compared with rural cohorts,[20] with exposure to livestock identified as a potential protective factor in the rural environment.[21] Convincing evidence supports that environmental factors, including those that differ between urban and rural areas, play a crucial role in shaping the immune response, which can affect the risk of developing allergies.

2.3 | Migration effects in the prevalence of food allergy

Australia has among the highest reported prevalences of childhood food allergy. Interestingly, a series of studies has shown that the prevalence of food allergy is around 3-fold higher among children born in Australia to parents who migrated from Asia, compared to children of Australian-born parents.[22–25] The parents themselves had a lower rate of allergy compared to Australian-born parents.[22] This increased risk of allergy in children born in Australia to Asian-born parents has been shown to a range of foods, including peanut, egg, tree nuts and cow's milk.[22] Conversely, children who were born in Asia and subsequently migrated to Australia had the lowest risk of food allergy.[26–28] In this study, introducing peanuts to a child before the age of 10 months was uncommon and had no discernible impact on the occurrence of peanut allergies. The increased risk of peanut allergy was also specifically observed in infants of Asian parents and was not found among infants of parents born in the UK or Europe. Altogether, these findings indicate that both genetic and environmental factors are likely to play a role in explaining this migration effect. However, the timing of introduction of allergenic foods such as peanuts alone is unlikely to explain these findings as early peanut introduction was uncommon among infants of both migrant and Australian-born parents.[22] Subsequent studies also showed a much

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TABLE 1 Time trends of food allergy prevalence.

Country	Reference	Location	Setting (urban/rural)	Year	Ethnicity	Age	No. of subjects	Methods of diagnosis (Hx, Hx + IgE, OFC)	Overall food allergy, %	Egg allergy, %	Peanut allergy, %
Time-trend studies on food allergy prevalence											
Australia	Soriano et al. 2022	Melbourne, Australia	Urban	2007–2011; 2018–2019	Australian-born (59%/49%); East Asian-born (11%/17%); Other-born parents (30%/34%)	12.4 (12.2–12.9) months; 12.5 (12.2–13.0) months	5276; 1933	Hx, SPT	NA	NA	2007–2011: 3.1% (95% CI, 2.7%–3.6%); 2018–2019: adj 2.6% (1.9–3.4)
China	Chen et al. 2011	Chongqing	Rural/urban	May 1999–Jun 1999	N/A	0–24 months	314	Qn + SPT ± OFC	OFC-confirmed FA 3.5%	2.90%	N/A [peanut sensitization: 1999: 1 (0.3%)]
	Hu et al. 2010			Jun 2009–Jul 2009			401		OFC-confirmed FA 7.7%	5.00%	N/A [peanut sensitization: 2009: 6 (1.5%)]
	Ma et al. 2021			Jun 2019–Aug 2019			513		OFC-confirmed FA 11.1%	5.50%	N/A [peanut sensitization: 2019: 2 (0.4%)]
Thailand	Rangkakulnutwat et al. 2021	Muang District, Chiang Mai Province	Rural	2010	N/A	3–7 years	546	Qn + SPT/sIgE + OFC	OFC-confirmed FA 1.1%; parent-reported FA 9.3%	NA	NA
			Rural	2019			600		OFC-confirmed FA 0.9%; parent-reported FA 5.5%	Parent-reported egg allergy 12.9%	Parent-reported peanut allergy 0%
UK	Venter et al. 2016	Isle of Wight	Urban	FAIR cohort: 2001–2002 IOW cohort: 1989	N/A	10 years	1456	Qn, SPT	FAIR: At 10 years: 3.6%, 2.54–5.15 IOW: at 10 years: 2.3%, 1.7–3.3%		FAIR: At 3–4 years: 1.2% At 10 years: 1.5% IOW: At 3–4 years: 0.62% At 10 years: 0.58%
USA	Gupta et al. 2011	USA	Urban	June 2009–Feb 2010	56.4% white, non-Hispanic; 21.6% Hispanic; 14.1% black, non-Hispanic; and 4.8% Asian, non-Hispanic	<18 years old, mean (CI) 8.5 (8.5–8.6) years	38,480	Parent-proxy Qn (Hx): 2/4 organ symptoms	8.0% (7.6%–8.3%)	0.8% (0.7%–0.9%)	2.0% (1.8%–2.2%)
USA	Gupta et al. 2018	USA	Urban	Oct 2015–Sep 2016	White, non-Hispanic (52.8%); Hispanic (24.1%); African American, non-Hispanic (13.2%); Asian Americans, non-Hispanic (3.2%)	<18 years old	38,408	Parent-proxy Qn (Hx): 2/4 organ symptoms	7.6% (7.1%–8.1%)	0.9% (0.7%–1.1%)	2.2% (2.0%–2.5%)

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higher rate of both early onset eczema and food allergy among Asian infants living in Australia compared to those living in Singapore, despite delayed introduction of allergenic foods among infants in Singapore.[29]

Although relatively limited to date, there is also consistent evidence indicating a higher prevalence of food allergy among Asian individuals living in other Western countries. An early study by Gupta and Colleagues (2011) reported that the likelihood of having a food allergy was significantly higher among Asian children compared to white children (OR 1.4 [95% CI, 1.2–1.7]), however, Asian children had significantly lower odds of having the allergy diagnosed (OR 0.7 [95% CI, 0.4–0.9]).[30] More recently, research has shown Asian children have somewhat higher rates of food allergy (6.2% vs 5.5%, OR 1.2 [95% CI, 1.1–1.4]) than non-Hispanic white children.[31] Research exploring food allergy in Asian-Indian-born immigrants to the United States found more than half of adult participants (57.6%, 95% CI, 50.7–64.3) reported food allergy onset during childhood or adolescence. Tree nut was the most commonly reported allergen (51.0% [95% CI, 46.8–55.2]), and over half (53.4% [95% CI, 49.2–57.5]) reported having more than one food allergy. Food-induced allergic reactions were reported by 82.3% (95% CI, 78.9–85.3) and anaphylaxis was reported in 21.7% (95% CI, 18.1–25.7) of those who had ever had a food-allergic reaction. Grandparents and parents of study participants had a significantly lower prevalence of food allergy compared with participants.[32]

The most recent study from the United States found that Asian (10.5% [95% CI, 9.1–12.0]), Hispanic (10.6% [95% CI, 9.7%–11.5%]), and non-Hispanic Black (10.6% [95% CI, 9.8%–11.5%]) individuals have high rates of food allergies compared to those of non-Hispanic white descent (9.5% [95% CI, 9.2–9.9]).[33] However, both Asian 46.9% (95% CI, 39.8%–54.1%) and non-Hispanic white individuals (47.8% [95% CI, 45.9–49.7]) had lower rates of severe food allergy compared to other races and ethnicities. Asian children reported higher rates of tree nut allergy (2.0% [95% CI, 1.2%–3.2%]) compared to other children, whereas Asian adults reported the highest prevalence of peanut allergy (2.9% [95% CI, 2.0–4.2]) and shellfish allergy (3.8% [95% CI, 3.0–4.9]). The analysis, however, did not account for the influence of birthplace and migration history.

In another population-based questionnaire survey, which showed a low prevalence of self-reported peanut allergy in Singapore overall (0.47%),[34] prevalence of peanut allergy was higher among Singapore expatriate children (1.2%) compared to local Singaporean children (0.47%–0.64%). However, the study did not investigate whether food allergies in the Singapore expatriate children pre-dated their arrival in Singapore.

Evidence from migration studies highlights the important role of epigenetic factors in shaping the risks of food allergies. The increased risk seen specifically in infants with Asian ancestry living in Westernized countries suggests their genetic background interacts with cultural environmental factors in a way that amplifies allergy development.[22,29] Unraveling these complex gene-environmental interactions is essential in developing more targeted approaches to allergy prevention and management.

2.4 | Prevalence and pattern of food allergies across the Asia-Pacific region

Limited data was available regarding the prevalence of food allergies in the AP region. Most studies relied on self-reported data, such as those from Singapore,[34–37] Thailand,[36,38–40] the Philippines,[34,36] Japan[41] and Vietnam,[19] which may increase the likelihood of overestimation. Very few studies included oral food challenges[39,40,42,43] and some included sIgE and/or SPT.[18,35] Table 2 summarizes the prevalence of food allergy in the AP region.

In the AP region, eggs were consistently reported as a common trigger for food allergies among young children across the region,[19,35,37,38] while shellfish prevailed among older children.[19,34,36] Nevertheless, there is a significant variation in the occurrence of certain food allergens in Asia. Wheat allergies were common in Thailand and Japan but relatively uncommon in other Asian regions.[44] Peanut and tree nut allergies are found to be more common in Westernized countries but remain low in the developing regions of Asia.[34] The prevalence of peanut allergy was low, ranging from 0% to 0.1% across Shaoguan, Guangzhou, and HK. The most prevalent food allergen in China was shrimp, whereas fish was found to be the most common allergen in Russia. In Vietnam, shellfish allergy was the major food allergen while peanut allergy was rare in both urban and rural areas of Vietnam (0.27% in Hue city and 0.31% in Tien Giang Province).[19] It is unclear why peanuts are uncommon food allergens in Asia, while they are consistently the top food allergens and common anaphylaxis triggers in the West.[45,46]

Findings are corroborated by a recent prospective study evaluating the anaphylaxis incidence and pattern across South/Southeast Asia between 2019 and 2022. The incidence of anaphylaxis was found to vary greatly between Qingdao (1 per 100,000 persons-year) and HK (11.09 per 100,000 persons-year) in China.[47] Among school-age children in HK and Singapore, peanuts were the top triggers of anaphylaxis, while peanut anaphylaxis was absent in Qingdao and very uncommon in Thailand. Despite the overall low prevalence of peanut allergies in Singapore and HK,[18,35] it is the leading cause of food-induced anaphylaxis in older children in this registry. This could be due to peanuts being a hidden ingredient in many Asian dishes. Additionally, peanut allergies may be more likely than egg and milk allergies to result in anaphylaxis.[48] Shellfish-induced anaphylaxis was more common in Thailand, but relatively less common in other regions of Asia, particularly Qingdao. In the past decade, bird's nest was a common cause of food-induced anaphylaxis in Singapore,[49] and anaphylaxis from ingesting insects was also reported in Thailand.[50,51] However, in this latest study, no anaphylaxis cases triggered by bird's nests and insects were observed, reflecting in the change in the consumption patterns over time.

Moreover, the prevalence and types of food allergies have been observed to vary over time, even within a single country. Over the past

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TABLE 2 Prevalence data on food allergy, confirmed by oral food challenges, history with specific IgE test and history alone.

Country	Reference	Location	Setting (urban/rural)	Year	Ethni-city	Age
OFC-confirmed FA						
Australia	Peters et al. 2017	Melbourne	Urban	Sep 2007–Aug 2011	Australian, East Asians	Recruitment at 12 months and followed up at 4 years
China	Dai et al. 2020	Wenzhou	Urban	N/A	N/A	3–6 years; average age was 3.99 ± 0.824 years
Europe	Xepapa-daki et al. 2015	9 European countries (Iceland, UK, Germany, Poland, Lithuania, Spain, Italy, Greece, Netherlands)	Urban	Recruitment Oct 2005–Feb 2010	N/A	Birth till 24 months
Thailand	Lao-araya et al. 2011	Muang District, Chiang Mai, Thailand	Rural	2010	Thai	3–7 years
Thailand	Sripramong et al. 2019	Bangkok, Thailand	Urban	2011–2015	N/A	Median of 6.5 years (0.1–29 years)
UK	Venter et al. 2016	Isle of Wight	Urban	Recruitment 2001–2002, follow-up at 10 years	N/A	10 years
South Africa	Botha et al. 2018	Cape Town	Urban	Feb 2013–Dec 2016	Black African; mixed ancestry	26 (22–32) months
		(Mqanduli district of the Eastern Cape)	Rural			21 (17–28) months
Convincing history and sIgE-confirmed FA						
China	Li et al. 2019	Hong Kong	Urban	Sep 2009–Jun 2016	Chinese	8.6 ± 1.6
		Guangzhou	Urban		Chinese	9.3 ± 1.7
		Shaoguan	Rural		Chinese	10.4 ± 2.0
India		Bangalore, Karnataka	Rural and rural		Indian	8.8 ± 1.5
Russia		Tomsk	Rural and rural		Russian	8.9 ± 1.1
Europe	Lyons et al. 2020	Zurich (Switzerland), Madrid (Spain), Athens (Greece), Sofia (Bulgaria), Lodz (Poland), Vilnius (Lithuania), Reykjavik (Iceland), and Utrecht (The Netherlands)	Urban	2007–2009	N/A	7–10 years
Japan	Ebisawa et al. 2020	Japan	Urban	2011	Japanese	All age
Singapore	Tham et al. 2018	Singapore	Urban	Jun 2009–Sep 2010	Chinese, Malay, Indian	12–48 months

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No. of subjects	Methods of diagnosis (Hx, Hx + IgE, OFC)	Overall IgE-mediated food allergy, %	Egg allergy, %	Peanut allergy, %
5276	SPT + OFC at 14–18 months	Any challenge-confirmed FA: Age 1 year: 11.0 (10.1–11.9) Age 4 years: 3.8 (3.3–4.4)	Age 1 year: 9.5 (8.7–10.3) Age 4 years: 1.2 (0.9–1.6)	Age 1 year: 3.1 (2.7–3.6) Age 4 years: 1.9 (1.6–2.3); late onset peanut allergy 0.2%
4151	Qn + SPT/sIgE ± OFC	at least 0.84% (children who were very likely to have FA and who dropped out during the study were considered negative for FA)	eggs (37.14%)	peanuts (5.71%)
12,049 enrolled; 9336 followed up to 2 years		NA	Challenge-confirmed HEA in the first 2 years of life: raw incidence 0.84 (0.67–1.03); adjusted incidence 1.23 (0.98–1.51)	NA
452	Hx, SPT, sIgE OFC	Overall 1.1% Shellfish 0.9% Fish 0.2%	NA	NA
2678	History, OFC if positive SPT	Perceived FA 26.1%; OFC-confirmed FA 1.08%	Egg white 4/15 = 26.7%, Egg yolk 5/14 = 35.7%	0/4
969 (827 FU at 10 years)	History, SPT ± OFC	Cumulative FA incidence: 6.8 (95% CI: 5.2–8.4); At 10 years: 3.6 (95% CI: 2.54–5.15)	Cumulative egg allergy: 1.4%; At 10 years: 0.73%	Cumulative peanut allergy: 1.4% At 10 years: 1.5%
1185	SPT + Hx/OFC	2.5 (1.6–3.3)	Raw egg: 1.9 (1.1–2.7) Cooked egg: 0.8 (0.3–2.3)	0.8 (0.3–1.3)
398		0.5 (95% CI, 0.1 to 1.8)	NA	NA
6194	Probable food allergy as defined by reported symptoms and positive SPT/IgE (≥0.7 kU/L) to the specific food (% [95% CI]) in different centers	1.5 (1.23–1.84) %	0.20 (0.11–0.34) %	0.10 (0.05–0.21) %
5542		0.21 (0.12–0.38) %	0.04 (0.01–0.13) %	0.00 (0.00–0.07) %
5139		0.69 (0.50–0.95) %	0.00 (0.00–0.07) %	0.00 (0.00–0.07) %
5677		0.14 (0.07–0.27) %	0.05 (0.02–0.16) %	0.03 (0.01–0.12) %
12,997		0.87 (0.73–1.05) %	0.10 (0.06–0.17) %	0.08 (0.05–0.15) %
Phase I – 16,935; Phase 2–670; Phase 3–18	Phase I – Qn; Phase 2 – Qn + sIgE Phase 3 – DBPCFC	Self-reported FA from 6.5% to 24.6% for priority foods; Probable FA from 1.9% (0.8–3.5) in Reykjavik to 5.6% (3.6–8.1) in Lodz	Lodz 0.76 (0.16–1.90); Madrid 0.89 (0.12–2.46); Vilnius 0.44 (0.02–2.28); Utrecht 0.21 (0.00–0.97); Zurich 0.00 (0.00–0.49); Athens 0.85 (0.01–3.06); Reykjavik 0.74 (0.15–1.84)	Lodz 0.78 (0.16–1.92); Madrid 0.89 (0.12–2.46); Vilnius 0.00 (0.00–0.86); Utrecht 0.63 (0.09–1.72); Zurich 0.41 (0.00–1.56); Athens 0.28 (0.07–1.89); Reykjavik 0.52 (0.06–1.48)
2954	< 1 year: 34% 1 year: 20% 2 years: 10%	Cow's milk 22% Wheat 12%	Eggs: 39%	Peanut 5%
1152 – >992 at 6 m – >902 at 12 months – >769 at 18 months – >881 at 24 months – >855 at 36 months – >851 at 48 months	Modified ISAAC Qn (convincing hx of an IgE mediated reaction to a food product) + SPT	12 months: 2.9% (26/902); 18 months: 2.7% (21/769); 24 months: 1.6% (14/881); 36 months: 1.1% (9/855); 48 months: 1.5% (13/851)	0.35%–1.8%	0.1%–0.3%

(Continues)

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TABLE 2 (Continued)

Country	Reference	Location	Setting (urban/rural)	Year	Ethni-city	Age
UK	Venkataraman 2018 et al.	Isle of Wight	Urban	Recruit-ment from Jan 1989 till Feb 1990	N/A	Birth till 18 years
Diagnosis by history						
Japan	Yamamoto-Hanada et al., 2020	Japan (15 study areas)	Urban	January 2011–March 2014	Japanese	Birth cohort (from birth till 3 years old)
Philippines	Shek etl al. 2010	Philippines	Urban rural	August 2007–February 2008	Filipino	14–16 years
Philippines	Connett et al. 1012	Philippines	Urban/rural	August 2007–February 2008	Filipino	14–16 years
Singapore	Shek et al. 2010	Singapore	Urban	August 2007–February 2008	Singaporean	4–6 years
			Urban			14–16 years
Singapore	Lee et al. 2014	Singapore	Urban	2014	Singaporean	11 m-30 m
Thailand	Santadusit et al. 2005	Bangkok, Thailand	Urban	NA	Thai	3 months-6 years
Vietnam	Le et al., 2018	Khanh Hoa province and Ho Chi Minh City	Rural	March–December 2016	Vietnamese	16–50 years
Vietnam	Le et al., 2018	Hue City	Urban	2016	Vietnamese	2–6 years
		Cai Be District, Tien Giang Province	Rural			

two decades, Japan has undertaken comprehensive national surveys to ascertain the prevalence and triggers of immediate food allergies.[52] The most recent survey, conducted in 2023, amassed a significant dataset of 6677 reported instances of food allergies. The data reveal a significant increase in cases among children under 1 year of age (30.9%), with hen's egg (33.4%), cow's milk (18.6%), and tree nuts (13.5%) being the primary culprits. Early-onset tree nut allergies, particularly walnut, have now become more prevalent than wheat allergies.[53] This trend is corroborated by a regional survey in Aichi prefecture, which recorded a spike in tree nut-induced anaphylaxis cases, climbing from 6.0% in 2017 to 15.3% in 2019.[54] Tree nut is typically not a common first solid food introduced into the diet of infants. However, there are recent changes in recommendations regarding the introduction of allergenic solids in Japan. In 2017, the Japanese Society of Pediatric Allergy and Clinical Immunology (JSPACI) first revised its guidelines to recommend introducing boiled hen's egg under medical supervision from 6 months of age to infants with atopic dermatitis. In 2019, the Ministry of Health, Labour and Welfare (MHLW) further recommended starting weaning from 5 to 6 months of age.[55] Overall, these studies provide valuable insights into the evolving landscape of food allergies across different demographics and over time. This shift in the allergic pattern reflects either changes in dietary patterns or increased awareness and reporting of tree nut allergies. Cross-cultural comparisons of food allergy epidemiology are essential to allow for the identification of factors, specifically dietary factors, that may contribute to these differences.

3 | EPIDEMIOLOGICAL EVIDENCE ON THE ROLE OF DIET IN THE DEVELOPMENT OF ALLERGY

Urbanization, modern dietary patterns, and allergic diseases are intricately linked. Early epidemiological studies, the PARSIFAL, and the follow-up ALADDIN birth cohort demonstrated that anthroposophic lifestyles might lower the incidence of atopy in children.[56–58] It has been suggested that dietary factors common in

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No. of subjects	Methods of diagnosis (Hx, Hx + IgE, OFC)	Overall IgE-mediated food allergy, %	Egg allergy, %	Peanut allergy, %
1536 recruited; 1456 followed up	Qn, SPT	1 year: 5.3% 2 years: 4.4% 4 years: 5.0% 10 years: 2.3% 18 years: 4%	1 year: 1.1% 2 years: 1.3% 4 years: 1.4% 10 years: 0.6% 18 years: 0.3%	1 year: 0.1% 2 years: 0.2% 4 years: 0.5% 10 years: 0.4% 18 years: 1.0%
104,062	Qn	1 year: 7.6% 2 years: 6.7% 3 years 4.9%	1 year: 5.4%	NA
11,434	Hx	Shellfish: 5.1% Tree nut: 0.3%	NA	Peanut: 0.4%
11,434	Hx	Fish: 2.29%	NA	NA
4515	Hx	Shellfish: 1.2% Tree nut 0.3%	NA	Peanut: 0.6%
6498		Shellfish: 5.2% Tree nut 0.3%	NA	Peanut: 0.5%
2737	Hx	Overall: 5% Cow's milk 0.5%	Egg: 1.8%	NA
656	Hx	Shellfish: 0.3%	Egg: 0.2%	NA
1185 (web); 9039 (paper)	Hx	Doctor-diagnosed FA (5.7%; $p = .7795$, $\beta = .05$); IgE-mediated FA (5.8%; $p = .9590$, $\beta = .05$)		
4443	Hx	Self-reported FA: 9.75 (8.91–10.65); Doctor-diagnosed FA: 8.40 (7.62–9.25)	Self-reported egg allergy: 1.15 (0.87–1.51); Doctor-diagnosed egg allergy 0.95 (0.70–1.28)	Self-reported PA: 0.47 (0.31–0.72); Doctor-diagnosed PA: 0.27 (0.15–0.47)
4177		Self-reported FA: 7.90 (7.12–8.76); Doctor-diagnosed FA: 4.96 (4.34–5.66)	Self-reported egg allergy: 1.10 (0.83–1.47); Doctor-diagnosed egg allergy: 0.74 (0.52–1.05)	Self-reported PA: 0.36 (0.22–0.59); Doctor-diagnosed PA: 0.31 (0.18–0.53)

anthroposophic households, for instance, a diet rich in vegetables, organic foods, fish, butter in bread, and olive oil may confer protection against atopic sensitization. However, other aspects of anthroposophic lifestyles, such as the limited use of vaccinations and antibiotics, may also have contributed to the differences in allergic diseases between the study groups, in accordance with the "hygiene hypothesis," which suggests that exposure to infectious diseases may have a protective effect against allergies.[59] Furthermore, the study only demonstrated differences in atopic sensitization and not in the prevalence of food allergies.

The diet of anthroposophic families closely resembles the Mediterranean diet, which consists of a high plant-to-animal food ratio.[60,61] This diet has been linked to lower asthma and allergy symptoms, and early-life adherence to Mediterranean diets can protect children from these conditions.[62–66] The PIFA study in the UK found that an infant diet high in vegetables, fruits, and home-cooked foods is linked to a reduced risk of food allergies by the age of two.[67] Maternal intake of vegetables and yogurt was also associated with less allergic rhinitis, atopic dermatitis, asthma, and wheezing in offspring by the age of 4 years.[68] Other birth cohorts also suggest that a balanced maternal diet before conception[69] and during pregnancy may reduce some early childhood wheezing phenotypes.[70]

Although the direct link between maternal and early infant dietary patterns and the development of food allergies is still unclear, there is a growing interest in the impact of microbially-derived short-chain fatty acids on the development of allergic diseases. Short-chain fatty acids (SCFAs) are produced when complex dietary and endogenous carbohydrates are broken down.[71] A recent systematic review showed that SCFAs, especially acetate, propionate, and butyrate, produced in the first few years of life were associated with reduced development of atopic dermatitis, wheezing or asthma, and IgE-mediated food allergy.[72] A high-fiber diet was shown to increase the production of SCFAs in mice and was found to be protective against food allergies in a mouse model.[73] However, this link is less clearly established in studies involving humans.[74] To date, no clinical trials have been published using SCFAs as an intervention for allergic

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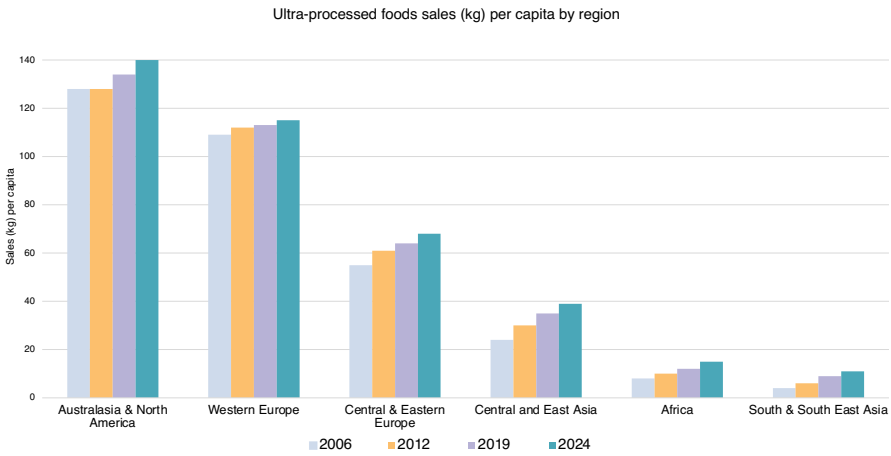

FIGURE 1 Sales of ultra-processed foods (UPFs) across different continents over time. The trend of UPF sales is increasing globally. Sales of UPFs in Australia and America are more than three times those in South and Southeast Asia. Figure was adopted from Baker et al.[116]

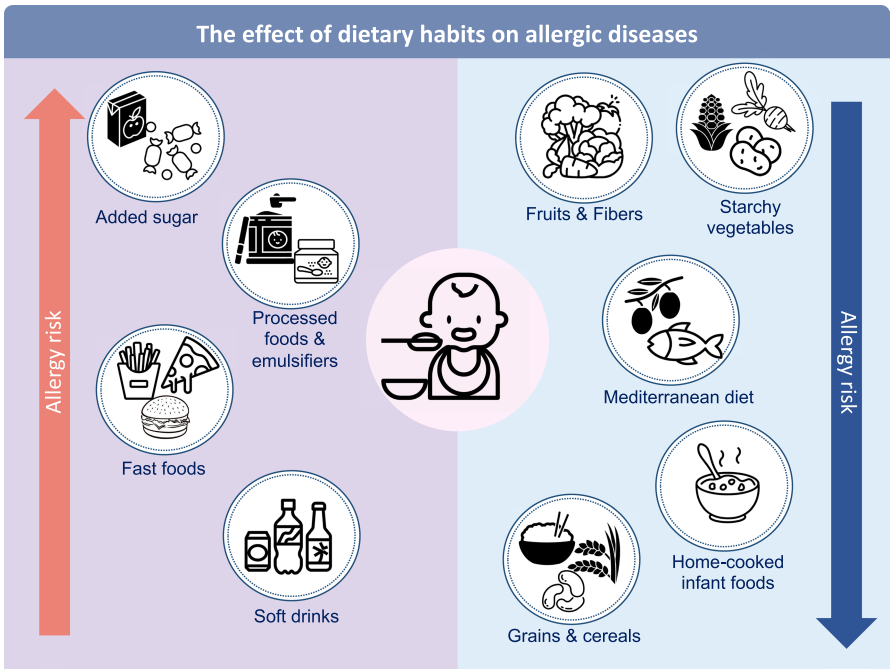

FIGURE 2 Summary of the dietary factors that may contribute to the development of allergy. The Mediterranean diet, which includes plenty of fruits and vegetables, fish and olive oil; a diet consisting of high fruits and fibers, starchy vegetables, cereals & grains, as well as home-cooked meals, have been shown to protect against allergies. Conversely, a diet rich in added sugar, fast and processed foods, and soft drinks has been shown to confer allergy risk.

diseases. Therefore, it is important to consider that the decrease in SCFAs observed among cases in previous studies may not represent the cause of the disease, but rather its effects, or may reflect other confounding factors. Further research is necessary to assess the potential of dietary interventions to modify the immune system for food allergy prevention and treatment.

4 | ANALYZING THE WORLDWIDE PHENOMENON OF NUTRITIONAL TRANSITION

4.1 | Global change in dietary preferences

In recent decades, food has been more widely available to the global population, both in terms of overall amount and diversity.[75] Economic growth facilitates international travel and cultural exchange, contributing to the diversification of local food landscapes. Research and guidelines also play an essential role in shaping our food preferences and dietary habits. Below we discuss the most noteworthy shift in dietary preferences.

4.1.1 | Peanuts

In Asia, peanuts are widely cultivated and consumed. China and India produce more than two-thirds of the world's peanuts, driven by their well-established peanut farming traditions, favorable climate and fertile agricultural lands.[76] China and India are consistently top peanut consumers, as most peanut consumption and oil production occurs in the regions where peanuts are primarily cultivated.[77,78] In fact, the consumption of peanuts is a quintessential culture in Asia and dates back to 1503.[79] Peanuts are traditionally boiled with salt, spices, and medicinal soup for over an

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hour. Boiling peanuts or preparing peanut-based dishes are common cultural practices in Vietnam (Lạc luộc), Philippines (nilagang mani), and Thailand (ถั่วลิสงต้ม). It has been demonstrated in vitro that boiling peanuts reduces their allergenicity, while roasting peanuts potentially increases their allergenic potential.[80–82] While the intentional introduction of peanuts into the diets of young children is generally uncommon, inadvertent exposure to peanuts at an early age is possible. Unlike Western cultures, where meals are usually served in separate portions with each person receiving their own plate of food, communal dining in Asian cultures is common and involves using shared chopsticks with meals in the middle of the table for everyone to share. It remains an intriguing paradox that in China, where environmental exposure to peanuts is high, the rate of peanut allergies remains relatively low compared to Western countries.

Peanuts, in contrast, have a relatively shorter history in the West. Peanuts were first consumed in the southern United States during the 1800s, and commercialization only occurred in the late 19th century due to advancements in transportation and technology. Peanut butter was first commercialized in 1928 and was quickly adopted by the soldiers as a staple diet during World War II. For centuries, peanuts were not commonly consumed. In a short period of time, the popularity of peanuts rapidly grew, as did the prevalence of peanut allergies. The frequency of peanut allergy in the United States reportedly increased 3.5-fold from 0.4% in 1997 to 0.8% in 2002 and 1.4% in 2008.[83] Since the LEAP study showed that introducing peanuts at an early age could prevent food allergies in UK infants,[84] many guidelines now recommend this practice to decrease the burden of immediate-onset peanut allergies.[85–87] However, the change in early peanut introduction practices accompanies a subtle shift in the pattern of food allergies, with a noticeable increase in peanut-induced FPIES in the United States[14,88] It appears that long-term exposure to an allergen may contribute to tolerance, while a sudden dietary shift may contribute to food allergy development.

The rate of peanut allergy in Singapore, however, remained low despite the delayed introduction of peanuts in the traditional Singaporean diet.[35] In Australia, early peanut introduction was linked to a decreased incidence of peanut allergy only in Australian-born infants, but not in East Asian infants.[11] This indicates that there is likely a complex interplay of genetic, environmental, and epigenetic factors contributing to food allergy risk. The timing of allergenic solids introduction is apparently not the only factor contributing to food allergies in Asians. Currently, the APAPARI does not recommend an early introduction of allergenic solids in Asian regions with a low incidence of food allergy/peanut allergy.[89]

4.1.2 | Wheat

Wheat products in Thailand and Japan have a relatively recent history, originating in the late 19th and early 20th centuries, despite being domesticated about 10,000 years ago.[90] Wheat is a cool-season crop and grows best in temperature climates (15-24°C) with specific moisture requirements at varying growth stages.[91] Extreme heat or frost can negatively impact wheat yields. In Thailand, where the temperature ranges from 25°C to 30°C, the majority of wheat is imported from overseas. Meanwhile, in Japan, the production of wheat is concentrated in the northern regions. As a result, a significant portion of the country's wheat supply still relies on imports.

Western influences and increased international trade have led to the widespread availability of wheat products. As Thailand and Japan experienced economic growth, wheat and wheat-based products, such as bread, pastries and pasta, have become integral to the Thai and Japanese diet alongside traditional rice-based dishes. In Thailand, wheat imports increased by more than seven-fold from 2000 to 2023.[92] Wheat-based infant cereal is becoming a common weaning diet for infants in urban regions of Thailand. The use of wheat-derived ingredients in cosmetic and skin care products has also gained popularity in Thailand and Japan in recent decades.[93] According to the dual-allergen exposure theory, early skin disturbance during infancy causes cutaneous sensitization and allergen penetration, which, when challenged orally, results in clinical allergies. On the other hand, early oral allergen exposure lowers allergy risk and increases tolerance.[94,95] Earlier reports showed a substantial correlation between the development of wheat allergy and exposure to hydrolysed wheat protein found in face soaps among Japanese women.[93,96] Accompanying the rapid increase in consumption of wheat and utilization of wheat-based products in the past century, the first case of wheat-dependent exercise-induced anaphylaxis was reported in Japan in 1985.[97] In the Asian anaphylaxis registries, wheat has also been confirmed as the primary cause of food-induced anaphylaxis among young children in Thailand and Japan, but not in most other Asian countries.[47,98]

4.1.3 | Tree nuts

Tree nut allergies have become more prevalent in developed regions of Asia in recent decades. Most notably, there has been an increase in nut consumption driven by a rise in health consciousness within Japan. Tree nuts are consumed not only in their whole form but also crushed, sliced, or processed into various products such as confectionery ingredients, milk, and oil. Since 2011, there has been a gradual increase in the annual consumption of walnuts as well as the imports of cashews and almonds in Japan.[99] Walnuts are particularly valued in East Asian countries due to their nutritional value, culinary versatility, and health benefits, especially with their alpha-linolenic acid, which is an omega-3 fatty acid beneficial for cognitive function development in children.[100] International trade has facilitated the widespread availability and affordability of walnuts in Japan, making them a convenient and encouraged snack and ingredient in the diets of Japanese children. Studies have indicated that environmental allergen levels of peanuts and walnuts are associated with sensitization.[101,102] Thus, the increase in domestic consumption and environmental level of tree nuts may be a contributing factor to the rise in tree nut allergies, but this is a complex issue that warrants further investigation.

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4.1.4 | Eggs

Eggs are a common food allergen among young children, and they are often introduced as the first solid food in infants' diets. Several large randomized controlled trials have demonstrated that early introduction of eggs into an infant's diet may help prevent the development of IgE-egg sensitization and egg allergy.[103] The PETIT study, conducted in Japan, further confirmed that infants with well-controlled eczema experienced the greatest benefits when introduced to cooked eggs.[104] As a result, the Allergology Society in Japan recommended the early introduction of boiled egg yolk to high-risk infants in 2017.[105] Between 2010–2017 and 2018–2019, there has been a 3.5-fold increase in the number of cases of egg-induced FPIES, with egg yolk being a more frequent FPIES trigger than egg white.[16] This observation suggests a potential correlation between early egg introduction and the emergence of this particular type of FPIES cases. Further prospective studies are necessary to evaluate this relationship.

4.2 | Regional change in dietary patterns

In relation to the economic expansion in developing nations, fresh meat, dry food items, and condiments are becoming more affordable, while the proportion of food sources from cultivating food crops continues to decrease. This phenomenon is known as the Bennett's Law, which is a development economics observation that individuals prefer to consume more nutrient-dense foods such meats, oils, sweeteners, and fewer calorie-dense starchy foods as their wealth rises.[106]

While maintaining some characteristics of their traditional food preparation and nutritional habits, most nations appeared to be moving towards being dominated by ultra-processed foods (UPFs). This pattern is often referred to as the "Western diet" as it is widespread among substantial populations in the United States, the United Kingdom, and Europe.[107] A modern or Western diet is characterized by high intakes of processed foods, including UPFs, refined grains and foods of animal origin, and low intakes of fruits, vegetables, and nuts.[7,108,109] UPFs currently account for almost half of the calories consumed in the United States,[110–112] the United Kingdom,[113] Canada,[114] and Australia.[115] Statistics from 2013 indicated that Japan ranked tenth in the retail sales of UPFs out of 80 countries evaluated. Although the UPF markets in Asia are also expanding, the growth rate is the lowest in South and Southeast Asia (Figure 1).[117–119] Below we discuss recent global dietary changes. A summary of changes is shown in Figure 2.

4.2.1 | China

Food supply in China was still a major problem in the 1970s, owing to restrictions in food trade and low availability of processed foods, thus farming was still the dominant occupation. Over the period of 10 to 20 years, certain regions in China have undergone a major shift towards becoming a modern, industrialized economy. However, significant income inequality remains, driven by urban–rural divide and regional economic variation.[120] Official reports indicate that 35% of China's population still resides in rural areas in 2022, compared to 64% in 2000,[121] and that the income difference between urban and rural areas is higher in western China than in eastern or central China.[122] Due to significant regional economic differences, it is crucial to evaluate the epidemiology of allergic diseases in China with care, as the prevalence of allergic diseases in one area cannot represent that of the entire country.

In the more modern part of the country, the Chinese diet is undergoing a significant change over time. Traditionally, it was rich in cereals and vegetables with low levels of fat and energy density. Dietary recalls have also shown that the traditional Chinese diet consisted of a high amount of protein-rich foods such as fungi, algae, legumes and seafoods.[123] However, since 1979, when major economic reforms took place, the diet has shifted to include more animal-based foods and decreased carbohydrate content.[124] Edible oil consumption gained traction, and obesity emerged in the 1980s. In major cities in China (Beijing, Shanghai, Xi'an, and Nanjing), about half of the children reported having consumed fast food over the past 3 months.[125] As children grew older, their fast food consumption increased by 12% each year. In this study, more than one in 10 children were obese and had hypertension. The surge in the consumption of processed foods and soft drinks parallels the rise in food allergy prevalence reported in Chongqing from 1999 to 2019, a major Chinese city that has undergone rapid economic development since 1997.[9] Modern China has swiftly progressed from a point where growth and nutrition were key concerns to a point where most parts of the nation are dealing with issues related to nutrition-related noncommunicable diseases such as obesity, metabolic and inflammatory disorders.[126,127] In contrast, high-income regions, such as the United States and Australia, have already experienced similar transitions between the 1990s and 2000s, at least 15 years earlier than the rise in food allergy prevalence observed in urban Chinese cities.

4.2.2 | Thailand

Thailand has seen a similarly dramatic increase in urbanization since the year 2000. The percentage of urban populations in the nation rose from 29% in 1990 to almost 50% in 2014.[128] Although Thailand also experienced notable economic growth, the pace was somewhat slower compared to China, and the growth rate was also more volatile. At present, a wide disparity still exists between individuals of high and low socioeconomic status. In 2017, a national survey on food consumption was conducted among a population aged 6 years and older, encompassing 27,960 families.[129] Two-thirds of individuals (70.5%) had not consumed Western fast food, while processed foods were consumed by 47% of individuals only, mostly one to two times per week, and uncommonly consumed on a daily basis (1.9%). The most common food groups that the Thai populations consume every day include fresh fruit (41%), eggs (15%), milk products (13%),

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fish (9%), and meat (7%). Even though the urbanized Thai regions have increasingly adopted lifestyle and food consumption behaviors akin to those in the Western world, the general Thai population, particularly those living in the rural regions, experienced minimal deviation from Eastern traditions.

Of note, Chiang Mai, a city in northern Thailand, has experienced gradual economic growth and urbanization in recent years, parallel to its level of air pollution. Seasonal smog and haze episodes due to agricultural burning and forest fires, along with increased urbanization and transportation demands, have contributed to air pollution challenges in this region. In spite of that, compared to Bangkok, where only 10% of its land is dedicated to agriculture, Chiang Mai's agricultural land usage of 43% is relatively high.[130] Supported by fertile soil and a favorable climate, Chiang Mai continues to emerge as a leader in organic and sustainable agricultural practices while expanding its agrotourism through the preservation of its rural heritage. The preserved farm environment and high intake of fresh produce likely play a role in maintaining a relatively stable time-trend food allergy prevalence.[39]

4.2.3 | Japan

Since the period of rapid economic growth, Japan has experienced a notable transformation in dietary preferences. Traditionally, the Japanese diet predominantly consisted of rice, fish, and vegetables. During the 19th and early 20th centuries, Japan experienced an influx of Western culture and dietary practices. The annual per capita rice consumption peaked at 118.3 kilograms in 1962 and has since steadily declined to approximately 51 kilograms as of 2021.[131] The diversification of diet has become more obvious since the mid-20th century. The annual per capita consumption of meat has increased almost 4 times, while fats and oils consumption has doubled. Additionally, seafood intake has declined over the years, with meat consumption surpassing seafood intake by the late 2000s. The fast-paced lifestyle and demands of urban living in the latter half of the 20th century further resulted in the popularity of convenience foods and processed foods. The dietary shift in Japan in the 1990s and 2000s is more developed than in other parts of Asia. Concurrently, an observational trend has emerged, where this shift in dietary patterns has coincided with a significant increase in food allergies in Japan, similar to the trend observed in Western countries. However, the exact nature of the relationship between these factors requires further research to establish causality.

5 | CONCLUSION

Asia is the largest continent in terms of land area and population. It boasts a diverse range of countries, from highly developed nations such as Japan to rapidly developing economies like China and transitional countries like Thailand. Comparing dietary patterns in relation to food allergy prevalence between Asian and Western countries can shed light on how changing lifestyles, urbanization, and Westernization of diet affect disease patterns. Such insights can aid in the development of targeted interventions and public health strategies. For example, implementing urban planning policies to support urban agriculture and local food markets can provide access to affordable, locally sourced food options in urban areas. Providing subsidies for small-scale farmers and taxes on unhealthy processed foods can be beneficial too.[132]

Evidence on the rural–urban differences and migration effects on food allergy clearly shows that environmental factors are the primary determinant of allergy risks. It is particularly important to understand why infants of Asian heritage living in a Westernized environment are more susceptible to food allergies. As transitional countries increasingly adopt a Western lifestyle and diet, a parallel rise in allergy prevalence may be expected, and it will be critical to monitor changes in prevalence in these countries. The historic dietary pattern in the East, which contrasts sharply with that of the West, also sheds light on several potential protective dietary features, which include the predominance of home-cooked and minimally processed foods, high carbohydrates and high-fiber diets. The observed increase in the occurrence of certain food allergies has been linked to changes in dietary habits, such as the rise of wheat and tree nut allergies in Asia in contrast to the rise in peanut allergies in the West. The recent surge in cases of egg and peanut-induced FPIES coincides with the new guideline change recommending earlier introduction of boiled eggs and peanuts. The shifting prevalence of tree nuts and wheat allergies, as well as egg and peanut-induced FPIES, calls for ongoing vigilance, research, and public health strategies to mitigate its impact. Further research is urgently needed to evaluate the effects of various dietary patterns on the risks of allergic diseases.

AUTHOR CONTRIBUTIONS

Agnes Sze-Yin Leung: Conceptualization; writing – original draft; writing – review and editing; data curation; project administration; methodology. **Punchama Pacharn:** Writing – original draft; writing – review and editing. **Supaluk Tangvalelerd:** Writing – original draft; writing – review and editing. **Sakura Sato:** Writing – original draft; writing – review and editing. **Erin Pitt:** Writing – original draft; writing – review and editing. **Gary Wong:** Writing – original draft; writing – review and editing. **Jennifer J. Koplin:** Writing – original draft; writing – review and editing; conceptualization; project administration.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest.

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